Asynchronous Max-Consensus Protocol with Time Delays: Convergence Results and Applications

Silvia Giannini, Antonio Petitti, *Student Member, IEEE,* Donato Di Paola, *Member, IEEE,* and Alessandro Rizzo, *Senior Member, IEEE*

***Abstract*—This paper deals with the analysis of the convergence properties of the max-consensus protocol in presence of asynchronous updates and bounded time delays on directed static networks. The work is motivated by real-world applications in distributed decision-making systems, for which max-consensus is an effective paradigm. The main result of this paper is that the strongly connectedness of the directed communication network is a sufficient condition for the asynchronous max-consensus protocol to let a distributed system converge in finite time. Implementation issues are also taken into account, by complementing the theoretical analysis with the definition of a mechanism to detect convergence in a distributed fashion. Finally, a numerical example is given, highlighting both the issues related to the failure of synchronous protocols applied to asynchronous settings and the effectiveness of the proposed asynchronous framework.**

***Index Terms*—Asynchronous protocols, max-consensus, multi-agent systems, networks.**

I. INTRODUCTION

CONSENSUS and agreement protocols are among the hottest research topics in the field of modeling and control of networked systems [1], [2], [3]. Consensus protocols allow all the nodes of a network to agree on a common value of an information of interest, such as a measurement, an action, or an opinion, only through local computation and communication [4], [5], [6]. Their application has been beneficial to many fields, such as sensor networks, wireless networks, and pools of unmanned vehicles, to name a few [7], [8], [9]. Among the several consensus protocols defined in the literature, max-consensus allows the nodes of a network to converge in finite time to the maximum of the initial values of their state variables [10]. Its applications mainly reside in the field of distributed-decision problems, such as leader election [11], task assignment [12], and distributed estimation [7], [13], [14], [15].

Much work in the literature assumes time synchronization of the network nodes' operations as a fundamental requirement for the correct design and implementation of consensus protocols. This scenario is very often far from reality, where at least drifts and offsets in nodes' clock ticks are present, up to cases involving heterogeneity in the network nodes and unreliable communication channels. In these and many other cases, asynchronism should be explicitly taken into account. Due to strict constraints on bandwidth and energy consumption that characterize distributed systems applications, the adoption of suitable time synchronization strategies may not be a feasible solution [16], [17] and the convergence properties of consensus protocols should be revisited in presence of asynchronous updates. In fact, the application of strategies conceived for synchronous systems to asynchronous frameworks may hamper the correct functionality of the protocol [18].

Relevant work in the literature addresses the convergence properties of the asynchronous average consensus protocol [19], [20], [21]. However, only little work has been devoted to an analogous analysis for the max-consensus protocol. In particular, convergence results have only been provided for synchronous switching topologies [22] and for probabilistic asynchronous frameworks [23].

This paper deals with the definition of an asynchronous discrete-time max-consensus protocol and the study of its convergence properties. The proposed asynchronous framework encompasses different scenarios: (i) synchronous systems with non uniform communication delays that make the delivery of information asynchronous; (ii) networks of nodes with different and/or not synchronized clock periods; (iii) a combination of the previous cases. Moreover, the framework presented in this paper allows nodes' clocks to have non-constant periods, which makes the framework suitable for modeling event-driven systems. To the best of our knowledge, this is the first work dealing with the analysis of the asynchronous max-consensus protocol in an extremely general framework. Convergence is proved under two assumptions: (i) communication delays cannot be arbitrarily long (*i.e.,* the so-called partial-asynchronism assumption [24]); and (ii) the underlying communication network is strongly connected. The convergence analysis is dealt with by establishing an equivalence between the proposed asynchronous protocol and

S. Giannini is with the Dipartimento di Ingegneria Elettrica e dell'Informazione (DEI), Politecnico di Bari, 70125 Bari, Italy, silvia.giannini@poliba.it, and with Selex ES, A Finmeccanica Company, 34077 Ronchi dei Legionari (GO), Italy.

A. Petitti is with the Institute of Intelligent Systems for Automation, National Research Council (ISSIA-CNR), 70126 Bari, Italy, petitti@ba.issia.cnr.it.

D. Di Paola is with the Institute of Intelligent Systems for Automation, National Research Council (ISSIA-CNR), 70126 Bari, Italy, dipaola@ba.issia.cnr.it.

A. Rizzo is with the Department of Mechanical and Aerospace Engineering, NYU Tandon School of Engineering, Brooklyn, NY 11201, USA, alessandro.rizzo@nyu.edu, and with the Dipartimento di Automatica e Informatica (DAUIN), Politecnico di Torino, 10129 Torino, Italy, alessandro.rizzo@polito.it.

This work was partially supported by the PON project (PON01_00980) project BAITAH "methodology and instruments of Building Automation and Information Technology for pervasive model of treatment and Aids for domestic Health-care", funded by the Italian MIUR.

Dr. Alessandro Rizzo warmly thanks the Honors Center of Italian Universities (H2CU) for housing and scholarship support.

Manuscript received May 22, 2015.

a synchronous protocol running on an ancillary switching topology, leveraging the formalism of analytic synchronization [25] and convergence results obtained for synchronous settings [26].

The results presented in this paper are grounded on two earlier conference publications by the same authors [13], [18], and extend them to the asynchronous max-consensus over directed networks with bounded non-uniform time delays and time-varying clocks over the network nodes.

The rest of the paper is structured as follows. Section II provides the necessary background. The asynchronous max-consensus protocol, along with its equivalent synchronous model, is defined in Section III. The convergence analysis is dealt with in Section IV, along with implementation issues and the definition of an event-based convergence detection strategy. Numerical results are given in Section V, followed by our conclusions in Section VI.

II. Background

A. Node and Network Model

Consider a network of $n > 1$ discrete-time dynamical systems, with index set $\mathcal{I} = \{1, \ldots, n\}$, operating in an asynchronous communication framework. Each node updates and transmits its state at time instants marked by its own local clock. Notably, we denote with $t_k^{[i]}$, $k \in \mathbb{N}_0$, the k-th update time for the i-th node (also referred to as k-th transmission time).

The set of update times of node $i \in \mathcal{I}$ is indicated with $\mathcal{T}^{[i]} = \{t_0^{[i]}, t_1^{[i]}, t_2^{[i]}, \ldots, t_k^{[i]}, \ldots\}$, where $t_0^{[i]}$ is the i-th node's starting time. No assumptions are made on the synchronization of starting times over the network. Therefore, the state evolution of each node must be referred to its local discrete time base, and a partial order among the update times of all the network nodes cannot be set without referring to an external, common time base.

The information state of node i at time $t_k^{[i]}$ is indicated with $\mathbf{x}^{[i]}(t_k^{[i]}) \in \mathbb{R}^m$. For simplicity of notation, in the following we always consider $m = 1$. This assumption does not compromise our proof of convergence for higher-dimension state vectors, provided that atomic operations are implemented and a proper max-operator for multidimensional vectors is defined. The application of atomic operations yields all the elements of the state vector of a given node to be updated and transmitted (resp. received) at the same time instant. A proper, multidimensional max-operator can be defined either as element-wise or to a norm over the state vector, thus reconducting the discussion to the case of max-consensus on scalar variables.

The communication network is modeled through a static directed graph (digraph) $\mathcal{G} = (\mathcal{I}, \mathcal{E})$, where the n dynamical systems constitute the node set $\mathcal{I}$, and the communication channels are modeled by the link set $\mathcal{E} \subseteq \mathcal{I} \times \mathcal{I}$. If node i is able to receive information by node j, then the ordered pair (j, i) belongs to $\mathcal{E}$, yet not necessarily vice versa. Self-loops are included in the graph, *i.e.,* $(i, i) \in \mathcal{E}$, $\forall i \in \mathcal{I}$. The set of one-hop incoming neighboring nodes of node i is defined as $\mathcal{N}^{[i]} = \{j \in \mathcal{I} | (j, i) \in \mathcal{E}\}$.

Each node $i \in \mathcal{I}$ asynchronously collects state values received from neighbors $j \in \mathcal{N}^{[i]}$ in a memory $\underline{\mathbf{x}}^{[i]} \in \mathbb{R}^{|\mathcal{N}^{[i]}| \cdot m}$. The updating mechanism of the memory will be explained in Sec. III. Initial conditions of the memory vector of node $i \in \mathcal{I}$ are set as $\underline{x}_j^{[i]}(t_{-\infty}^{[i]}) = -\infty$, $\forall j \in \mathcal{N}^{[i]}$, $j \neq i$, and $\underline{x}_i^{[i]}(t_{-\infty}^{[i]}) = x^{[i]}(t_0^{[i]})$. Finally, the following two assumptions are formulated.

Assumption II.1 (Bounded communication delays)**.** *Let us define $\theta_k^{ij} \in \mathbb{R}_+$ the time at which the information transmitted by node $j \in \mathcal{N}^{[i]}$ at some update time $t_q^{[j]}$, $q \in \mathbb{N}_0$ is received by node i in the reception interval $[t_k, t_{k+1})$ (here, θ_k^{ij} is measured with reference to $t_0^{[i]}$). It is assumed that, for each node $i \in \mathcal{I}$, and for each delivery time θ_k^{ij}, $j \in \mathcal{N}^{[i]}$, $k \in \mathbb{N}_0$, the constraint $0 \leq d_k^{ij} \leq \check{d}$, $\check{d} \in \mathbb{R}_+$, holds, where d_k^{ij} is the delivery delay,* i.e., *the time interval elapsed between $t_q^{[j]}$ and θ_k^{ij}. It results in $\theta_k^{ii} = t_k^{[i]}$ and $d_k^{ii} = 0$, $\forall i \in \mathcal{I}$, $k \in \mathbb{N}_0$.*

Assumption II.2 (Strong connectivity)**.** *Digraph $\mathcal{G}$ is assumed to be strongly connected, that is, there always exists a directed path between any two nodes of the digraph [27].*

A path in the network is an ordered sequence of links connecting two arbitrary nodes. The length of a path is the number of links in the path. In a strongly connected digraph, the longest of the shortest paths connecting any two nodes is the graph diameter, indicated with $\mathcal{D}$ [27]. The adjacency matrix $\mathcal{A} = [a^{ij}]$ of the digraph $\mathcal{G}$ is defined as

$$a^{ij} = \begin{cases} 1 & \text{if } (j, i) \in \mathcal{E} \\ 0 & \text{otherwise} \end{cases}, \tag{1}$$

and $a^{ii} = 1$, $\forall i \in \mathcal{I}$, is assumed.

A digraph $\mathcal{G}$ is called static if the edge set $\mathcal{E}$ is time-invariant. On the other hand, a dynamic digraph $\mathcal{G}_k = (\mathcal{I}, \mathcal{E}_k)$, $k \in \mathbb{N}_0$, is a digraph with a time-varying edge set. Dynamic digraphs are useful to model communication structures that change over times and can be consistently described by time-varying adjacency matrices. A set of dynamic digraphs $\mathcal{G}_1, ..., \mathcal{G}_r$, $r \in \mathbb{N}$, is jointly strongly connected if the union digraph $\mathcal{U} = \bigcup_{p=1}^{r} \mathcal{G}_p$, with vertex and edge set defined as $\mathcal{U} = (\mathcal{I}, \bigcup_{p=1}^{r} \mathcal{E}_p)$, is strongly connected.

B. Synchronous Max-Consensus Protocol

Consider a set of $n > 1$ discrete-time dynamical systems with synchronized update times, *i.e.,* $t_k^{[i]} = t_k, \forall i \in \mathcal{I}, k \in \mathbb{N}_0$, that communicate over a static digraph $\mathcal{G} = (\mathcal{I}, \mathcal{E})$ with no delays (*i.e.,* $d_k^{ij} = 0$, $\forall (j, i) \in \mathcal{E}$, $k \in \mathbb{N}_0$). The scalar[1] state variable of each system is indicated with $\widetilde{x}^{[i]} \in \mathbb{R}$, $i \in \mathcal{I}$. Suppose that each node runs the synchronous max-consensus protocol $\widetilde{\mathcal{P}}$ defined as

$$\widetilde{x}^{[i]}(t_{k+1}) = \max_{j \in \mathcal{N}^{[i]}} \{\widetilde{x}^{[j]}(t_k)\}, \quad i \in \mathcal{I}. \tag{2}$$

[1] Similarly to the asynchronous case, the state variable can be easily replaced by a state vector $\widetilde{\mathbf{x}}^{[i]} \in \mathbb{R}^m$.

Theorem II.1 (Convergence property of the synchronous max-consensus protocol)**.** *If $\mathcal{G}$ is strongly connected, then*

$$\begin{aligned}\widetilde{x}^{[i]}(t_k) &= \widetilde{x}^{[j]}(t_k)\\ &= \max\Big(\widetilde{x}^{[1]}(t_0), \cdots, \widetilde{x}^{[n]}(t_0)\Big)\ \forall i, j \in \mathcal{I},\ \forall k \geq \mathcal{D}.\end{aligned}$$

Proof. The proof of Theorem II.1 is reported in [10]. □

We remark that the results presented in this paper for the max-consensus protocol can also be applied to the min-consensus problem, which can be formulated as a max-consensus one where node state values are multiplied by -1.

III. Asynchronous Max-Consensus: Equivalent Synchronous Model

A. Asynchronous Max-Consensus Protocol

Suppose that node $i \in \mathcal{I}$ updates its information state at each own local update time $t_k^{[i]} \in \mathcal{T}^{[i]}$, $k \in \mathbb{N}$, on the basis of the value of its state variable at the previous update time $t_{k-1}^{[i]}$ and on the most recent values of the state variables received from the neighboring nodes, which are collected in the memory vector $\underline{\mathbf{x}}^{[i]}$. To this aim, we assume detectable delays (*e.g.,* through a sequence number) and the adoption of the most-recent-data strategy [28].

The asynchronous max-consensus protocol $\mathcal{P}$ on digraph $\mathcal{G}$, for a generic node $i = 1, \ldots, n$, is defined by the following state update rule:

$$x^{[i]}(t_{k+1}^{[i]}) = \max_{j \in \mathcal{N}^{[i]}} \{\underline{x}_j^{[i]}(t_k^{[i]})\}, \quad t_{k+1}^{[i]} \in \mathcal{T}^{[i]},\ k \in \mathbb{N}_0, \quad (3)$$

where the memory vector is locally updated as follows:

$$\begin{cases} \underline{x}_j^{[i]}(t_k^{[i]}) = x^{[j]}(t_q^{[j]}) & t_k^{[i]} \leq \theta_k^{ij} < t_{k+1}^{[i]} \quad (i)\\ \underline{x}_j^{[i]}(t_k^{[i]}) = \underline{x}_j^{[i]}(t_{k-1}^{[i]}) & \text{otherwise} \quad (ii)\end{cases}. \quad (4)$$

As stated in Assumption II.1, variable $\theta_k^{ij} \in \mathbb{R}_+$ indicates the time instant at which the most recent state value of the neighboring node $j \in \mathcal{N}^{[i]}$, available to node i before time $t_{k+1}^{[i]} \in \mathcal{T}^{[i]}$, is received by agent i. Thus, $t_0^{[i]} \leq \theta_k^{ij} < t_{k+1}^{[i]}$, $\forall j \in \mathcal{N}^{[i]}, k \in \mathbb{N}_0$, holds. Moreover, it follows that $\underline{x}_i^{[i]}(t_k^{[i]}) = x^{[i]}(t_k^{[i]})$, $\forall i \in \mathcal{I}, k \in \mathbb{N}_0$.

The asynchronous system under exam is defined by the triple

$$\mathcal{S} = \langle \mathcal{G}, \mathcal{P}, \mathcal{T} \rangle, \quad (5)$$

where $\mathcal{G}$ is the static digraph representing the communication structure, $\mathcal{P}$ is the asynchronous max-consensus protocol defined in Eq. (3), and $\mathcal{T}$ is the unordered set of all the update times of all the nodes in the network, *i.e.,* $\mathcal{T} = \bigcup_{i=1}^{n} \mathcal{T}^{[i]}$.

Problem III.1 (Asynchronous max-consensus problem)**.** *Given system $\mathcal{S}$ in (5), max-consensus is achieved in finite time if, under the execution of protocol $\mathcal{P}$ in (3), there exists a $\nu^{[i]} \in \mathbb{N}$, $\forall i \in \mathcal{I}$, s.t. $\forall k \geq \nu^{[i]}$, $k \in \mathbb{N}$, $x^{[i]}(t_k^{[i]}) = \max_{j \in \mathcal{I}}\{x^{[j]}(t_0^{[j]})\}$ holds.*

In the rest of the section, we show that system $\mathcal{S}$, which operates with the asynchronous update rules of protocol $\mathcal{P}$ in (3) over a static digraph $\mathcal{G} = (\mathcal{I}, \mathcal{E})$, is equivalent to a suitably defined system $\widetilde{\mathcal{S}}$ where a synchronous protocol $\widetilde{\mathcal{P}}$ is run on a switching topology $\widetilde{\mathcal{G}}_h = (\mathcal{I}, \mathcal{E}_h)$, $h \in \mathbb{N}_0$. Hence, Problem III.1 will be addressed by resolving an ancillary max-consensus problem (defined in this paper in Problem III.2) operating on the equivalent switching digraph.

B. Virtual Time Base

As stated in Section II, due to the very general assumptions made in this study, it is not possible to order the set of nodes' update times without referring them to an external, common time base, which is needed for analysis purposes. Following the line of arguments of the analytic synchronization method [24] [25], a global ordering of the time instants in the set $\mathcal{T} = \bigcup_{i=1}^{n} \mathcal{T}^{[i]}$ is defined by assuming the point of view of a hypothetical external observer of the network. Hence, let us denote with $\mathcal{T}^S$ the sorted version of the set $\mathcal{T}$. The common discrete time base is then formalized as

$$\Upsilon = \{\tau_h | \tau_h = h \cdot c^O, h \in \mathbb{N}_0, \tau_h \in \mathbb{R}_+\}, \quad (6)$$

where c^O is a constant clock period selected as the biggest time interval contained, an integer number of times, in every time interval occurring between two consecutive update times in the sorted set $\mathcal{T}^S$. A picture of the introduced time bases is reported in Fig. 1(b) with reference to the linear topology in Fig. 1(a). In the following, we will refer to Υ as to the virtual time base, since it is not actually possessed by any node, yet it is only used for analysis purposes. Thus, a correspondence between any of the update times of the state of the nodes and a specific virtual time instant can be set. Note that this correspondence is not a bijection, that is, each element of $\mathcal{T}^S$ corresponds to an element of Υ, whereas the opposite may not hold, *i.e.,* $|\mathcal{T}^S| \leq |\Upsilon|$ is verified. We formalize this relation by means of the following set of functions:

$$f^{[i]} : \mathcal{T}^{[i]} \to \Upsilon, \quad i \in \{1, \ldots n\}. \quad (7)$$

The expression $f^{[i]}(t_k^{[i]}) = \tau_{k_i}$, $k \in \mathbb{N}_0$, means that given $t_k^{[i]} \in \mathcal{T}^{[i]}$, with $\mathcal{T}^{[i]} \subseteq \mathcal{T}^S$ and $i \in \mathcal{I}$, $\tau_{k_i} \in \Upsilon$ represents its corresponding virtual update time. With a slight abuse of notation, $\tau_{k_i} \in \mathcal{T}^{[i]}$ denotes the existence of a virtual time instant $\tau_{k_i} \in \Upsilon$ and of a time instant $t_k^{[i]} \in \mathcal{T}^{[i]}$, such that $\tau_{k_i} = f^{[i]}(t_k^{[i]})$.

C. Equivalent Synchronous Model

Similarly to the virtual time base, also the switching structure can only be revealed by observing system $\mathcal{S}$ from an external, global point of view. In particular, the node sets of $\mathcal{G}$ and $\widetilde{\mathcal{G}}_h$ coincide for all $h \in \mathbb{N}_0$, whereas the time-varying edge sets of digraphs $\widetilde{\mathcal{G}}_h$ evolve as a function of the effective, possibly unidirectional, data exchange flows occurring between pairs of nodes during the virtual time windows $(\tau_{h-1}, \tau_h]$, $h \in \mathbb{N}$. Notably, a directed edge from a node $j \in \mathcal{N}^{[i]}$ to node i is switched on at τ_h if and only if $\tau_h = \tau_{k_i}$, with $\tau_{k_i} \in \mathcal{T}^{[i]}$. We assume that every node can always access its own state, that is, $(i, i) \in \mathcal{E}_h$, $\forall \tau_h \in \Upsilon$. Digraph $\widetilde{\mathcal{G}}_h$, which evolves along the virtual time base Υ, is

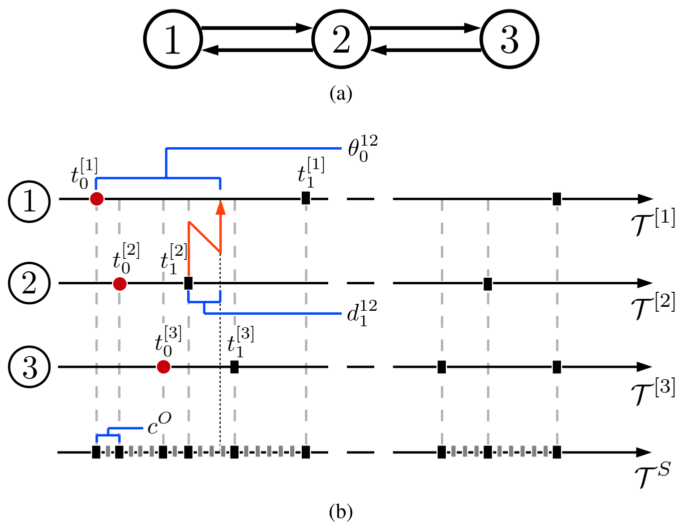


Fig. 1. Communication topology of a network of three nodes, with diameter $\mathcal{D} = 2$ (Fig. 1(a)), and their update times (Fig. 1(b)). In Fig. 1(b), the time instant θ_0^{12} at which the most recent state value of node 2 is received by node 1, the delivery delay d_1^{12}, *i.e.,* the time interval elapsed between $t_1^{[2]}$ and θ_0^{12}, and the constant clock c^O within the set of ordered time istants are also shown.

described by a time-varying adjacency matrix $\widetilde{A}_h \in \{0,1\}^{n\times n}$ whose elements $\widetilde{a}_h^{ij} \in \{0,1\}$ are defined as:

$$\widetilde{a}_h^{ij} = \begin{cases} a^{ij} & \text{if } \tau_h = \tau_{k_i},\ j \in \mathcal{N}^{[i]}\backslash\{i\} \\ 1 & i = j \\ 0 & \text{otherwise} \end{cases} . \tag{8}$$

The dynamic set of neighbors of node i at time $\tau_h \in \Upsilon$, $h \in \mathbb{N}_0$, is defined as

$$\widetilde{\mathcal{N}}_h^{[i]} = \{j \in \mathcal{I} | \widetilde{a}_h^{ij} = 1\}, \tag{9}$$

where $\widetilde{\mathcal{N}}_h^{[i]} \subseteq \mathcal{N}^{[i]}$, $\forall h \in \mathbb{N}_0$, holds.

Consider now the following synchronous max-consensus protocol $\widetilde{\mathcal{P}}$, running over the switching digraph $\widetilde{\mathcal{G}}_h$:

$$\widetilde{x}^{[i]}(\tau_{h+1}) = \max_{j\in\widetilde{\mathcal{N}}_h^{[i]}} \left\{ \underline{\widetilde{x}}_j^{[i]}(\tau_h) \right\}, \quad \tau_h \in \Upsilon, \tag{10}$$

where

$$\begin{cases} \underline{\widetilde{x}}_j^{[i]}(\tau_h) = \widetilde{x}^{[j]}(\tau_{q_j}) & \text{if } \tau_h - \tau_{q_j} \le d_h^{ij} < \tau_{h+1} - \tau_{q_j}, \quad (i) \\ \underline{\widetilde{x}}_j^{[i]}(\tau_h) = \underline{\widetilde{x}}_j^{[i]}(\tau_{h-1}) & \text{otherwise.} \quad (ii) \end{cases} \tag{11}$$

It results that $\underline{\widetilde{x}}_i^{[i]}(\tau_h) = \widetilde{x}^{[i]}(\tau_h)$, $\forall \tau_h \in \Upsilon, i \in \mathcal{I}$. Also in this case, the initial conditions of the memory vector are set as $\underline{\widetilde{x}}_j^{[i]}(\tau_{0_i}) = -\infty$, $\forall j \in \mathcal{N}^{[i]} - \{i\}$, while $\underline{\widetilde{x}}_i^{[i]}(\tau_{0_i}) = \widetilde{x}^{[i]}(\tau_{0_i})$.

According to protocol $\widetilde{\mathcal{P}}$ in (10), every node $i \in \mathcal{I}$ updates its state synchronously at time $\tau_h \in \Upsilon$. Differently from protocol $\mathcal{P}$ in (3), though, its neighbor set changes over the time instants τ_h. Hence, Eq. (10) can be rewritten as follows:

$$\begin{cases} \widetilde{x}^{[i]}(\tau_{h+1}) = \max_{j\in\mathcal{N}^{[i]}} \left\{ \underline{\widetilde{x}}_j^{[i]}(\tau_h) \right\} & \forall \tau_h \in \mathcal{T}^{[i]} \quad (i) \\ \widetilde{x}^{[i]}(\tau_{h+1}) = \widetilde{x}^{[i]}(\tau_h) & \forall \tau_h \notin \mathcal{T}^{[i]} \quad (ii) \end{cases} . \tag{12}$$

We define the synchronous system described above with the triple

$$\widetilde{\mathcal{S}} = \langle \widetilde{\mathcal{G}}, \widetilde{\mathcal{P}}, \Upsilon \rangle, \tag{13}$$

where the synchronous protocol $\widetilde{\mathcal{P}}$ in (10) is run over the switching topology $\widetilde{\mathcal{G}}$, described by the adjacency matrix (8), along the virtual time base Υ expressed by (6).

Problem III.2 (Synchronous max-consensus problem)**.** *Given system $\widetilde{\mathcal{S}}$, protocol $\widetilde{\mathcal{P}}$ solves the synchronous max-consensus problem if there exists an integer $\widetilde{\nu} \in \mathbb{N}$ s.t. $\forall \widetilde{k} \ge \widetilde{\nu}$, $\widetilde{k} \in \mathbb{N}$, $\widetilde{x}^{[i]}(\tau_{\widetilde{k}}) = \max_{j\in\mathcal{I}}\{\widetilde{x}^{[j]}(\tau_{0_j})\}$, $\forall i \in \mathcal{I}$, holds.*

We recall here a lemma on the convergence property of the synchronous max-consensus protocol over switching topologies, useful for subsequent analysis.

Lemma III.1. *Given a finite sequence of r adjacency matrices $\widetilde{A}_1, \dots, \widetilde{A}_k, \dots, \widetilde{A}_r$, synchronous system (13) achieves max-consensus in finite time, for all initial condition vectors $\widetilde{\mathbf{x}}_\mathbf{0} = [\widetilde{x}(\tau_{0_1}) \dots \widetilde{x}(\tau_{0_n})]^T$, if and only if the corresponding switching sequence of digraphs $\widetilde{\mathcal{G}}(\widetilde{A}_1), \dots, \widetilde{\mathcal{G}}(\widetilde{A}_k), \dots, \widetilde{\mathcal{G}}(\widetilde{A}_r)$ is jointly strongly connected.*

Proof. The proof of Lemma III.1 is reported in [26]. □

IV. Asynchronous Max-Consensus: Convergence

A. Convergence Analysis

Consider a networked system $\mathcal{S}$ characterized by triple (5), and its corresponding synchronous system $\widetilde{\mathcal{S}}$ expressed by triple (13).

Proposition IV.1. *If system $\widetilde{\mathcal{S}}$ in (13) reaches max-consensus in finite time, that is, if protocol $\widetilde{\mathcal{P}}$ solves the synchronous max-consensus problem III.2 over the switching network $\widetilde{\mathcal{G}}_h$, $h \in \mathbb{N}_0$, then system $\mathcal{S}$ in (5) reaches max-consensus in finite time, that is, protocol $\mathcal{P}$ solves the asynchronous max-consensus problem III.1 over the static network $\mathcal{G}$. Moreover, if the two state-variable sets $\{\widetilde{x}^{[i]}\}_{i\in\mathcal{I}}$ and $\{x^{[i]}\}_{i\in\mathcal{I}}$, of systems $\widetilde{\mathcal{S}}$ and $\mathcal{S}$ respectively, assume identical initial values,* i.e., $\widetilde{x}^{[i]}(\tau_{0_i}) = x^{[i]}(t_0^{[i]})$, $\forall i \in \mathcal{I}$, *then the two systems converge to the same value, that is,* $\max_{i\in\mathcal{I}}\{\widetilde{x}^{[i]}(\tau_{0_i})\} = \max_{i\in\mathcal{I}}\{x^{[i]}(t_0^{[i]})\}$.

Proof. The proof, inspired by the line of arguments presented in [28], is reported in Appendix to improve readability. □

Grounded on Proposition IV.1, we shall prove the convergence result of protocol $\mathcal{P}$ in $\mathcal{S}$, under the mild assumption of partial asynchronism, which will be defined below.

Definition IV.1. *(Partially asynchronous algorithm [24], [25]) An asynchronous algorithm is called partially asynchronous if there exists a positive constant $B \in \mathbb{N}$, called asynchronism measure, for which the following two conditions hold:*

1) *each node performs an update at least once in B time units;*
2) *the information used by any node is outdated by at most B time units.*

Assumption IV.1. *It is assumed that protocol $\mathcal{P}$ expressed by (3) is partially asynchronous, that is, local state updates* *cannot occur arbitrarily slow, with respect to the virtual time base.*

Thus, system $\mathcal{S}$, to which protocol $\mathcal{P}$ is applied, is also partially asynchronous with asynchronism measure B, and its partial asynchronism condition is specialized as

Assumption IV.2. *There exists a positive integer B s.t.*

1) *for every $i \in \mathcal{I}$ and for every $\tau_h \in \Upsilon$, $h \in \mathbb{N}_0$, there exists at least one value $k \in \mathbb{N}_0$ s.t. $f^{[i]}(t_k^{[i]}) \in \{\tau_h, \tau_{h+1}, ..., \tau_{h+B-1}\}$, with $t_k^{[i]} \in \mathcal{T}^{[i]}$;*
2) *defining $\tau_{q_j} \in \mathcal{T}^{[j]}$ as the virtual time at which node $j \in \mathcal{N}^{[i]}$ produces the information used by node i at time $\tau_{k_i} \in \mathcal{T}^{[i]}$ through the state update rule (3), there holds: $\tau_{k_i} - B \cdot c^O < \tau_{q_j} \le \tau_{k_i}$, $\forall (j,i) \in \mathcal{E}$, $i \neq j$, $\forall t_k^{[i]} \in \mathcal{T}^{[i]}$.*

Assumption IV.2 implies that each node updates its state at least once during any time interval of length $B \cdot c^O$, and that, given any node, the state information received by its neighbors is outdated by no more than B time units, expressed in the virtual time base. We observe that, under Definition IV.1, the synchronous case is a particular case of the asynchronous one, with asynchronism measure $B = 1$.

The proof of convergence for the asynchronous max-consensus protocol is based on the following lemma.

Lemma IV.1. *Given the partially asynchronous system $\mathcal{S} = \langle \mathcal{G}, \mathcal{P}, \mathcal{T} \rangle$ defined in (5) and its equivalent synchronous system $\widetilde{\mathcal{S}} = \langle \widetilde{\mathcal{G}}, \widetilde{\mathcal{P}}, \Upsilon \rangle$ defined in (13), if the digraph $\mathcal{G}$ is strongly connected, then $\forall \tau_h \in \Upsilon$, $\exists b_h \in \mathbb{N}$ s.t. the sequence $\{\widetilde{\mathcal{G}}_h, \dots, \widetilde{\mathcal{G}}_k, \dots, \widetilde{\mathcal{G}}_{h+b_h}\}$ is jointly strongly connected. Moreover, it results in $b_h \le B-1$, $\forall h \in \mathbb{N}_0$.*

Proof. Consider node $i \in \mathcal{I}$ and a virtual time $\tau_h \in \Upsilon$, $h \in \mathbb{N}_0$. The partial asynchronism assumption IV.2 guarantees that in at most B clock units of the virtual time base, node i receives the values of the state variables of every neighbor $j \in \mathcal{N}^{[i]}$, that is, $\bigcup_{p=h}^{h+B-1} \widetilde{\mathcal{N}}_p^{[i]} = \mathcal{N}^{[i]}$, $\forall i \in \mathcal{I}$. Analogously, for all $\tau_h \in \Upsilon$, it results in $\mathcal{G} = \bigcup_{p=h}^{h+B-1} \widetilde{\mathcal{G}}_p$ and $\mathcal{A} = \bigvee_{p=h}^{h+B-1} \widetilde{\mathcal{A}}_p$, where $\bigvee$ is the element-wise logical `or` of the adjacency matrices $\widetilde{\mathcal{A}}_p$, describing the switching digraphs $\widetilde{\mathcal{G}}_p$ at times $\tau_p \in \Upsilon$, $p \in \{h, \dots, h+B-1\}$. The lemma is proved under Assumption II.2 (*i.e.,* the strongly connectedness of the static digraph $\mathcal{G}$), since it is possible to consider $b_h = B-1$, $\forall \tau_h \in \Upsilon$. □

The main results of this paper can be summarized by the following convergence Theorem, which also provides an upper bound on the convergence time when the asynchronous max-consensus protocol is adopted.

Theorem IV.1. *Let $\mathcal{S} = \langle \mathcal{G}, \mathcal{P}, \mathcal{T} \rangle$ be the system in (5) running on protocol $\mathcal{P}$ in (3). If digraph $\mathcal{G}$ is strongly connected, then system $\mathcal{S}$ in (5) reaches convergence in finite time, solving Problem III.1, for all initial conditions $x^{[i]}(t_0^{[i]})$, $i \in \mathcal{I}$. Moreover, an upper bound on the convergence time is given by $B \cdot \mathcal{D}$ time steps, expressed in the virtual time base, where B is the asynchronism measure of $\mathcal{S}$, and $\mathcal{D}$ is the diameter of $\mathcal{G}$. More specifically, system $\mathcal{S}$ reaches max-consensus in at most $B \cdot \mathcal{D} \cdot c^O$ time units.*

Proof. The proof follows directly from the results stated in Lemmas III.1 and IV.1, and on the basis of the equivalence between systems $\mathcal{S}$ and $\widetilde{\mathcal{S}}$, stated in Proposition IV.1.
Consider the synchronous equivalent model $\widetilde{\mathcal{S}}$ of $\mathcal{S}$, and the following $\mathcal{D}$ sequences of B adjacency matrices: $S_1 \triangleq \widetilde{\mathcal{A}}_1, \dots, \widetilde{\mathcal{A}}_B$; $S_2 \triangleq \widetilde{\mathcal{A}}_{B+1}, \dots, \widetilde{\mathcal{A}}_{2B}$; $\dots$; $S_{\mathcal{D}} \triangleq \widetilde{\mathcal{A}}_{(\mathcal{D}-1)B+1}, \dots, \widetilde{\mathcal{A}}_{\mathcal{D}B}$. Each adjacency matrix models the network topology at the virtual time τ_h, $h \in \{1, \dots, \mathcal{D} \cdot B\}$. From Lemma IV.1, it follows that each sequence S_k, $k \in \{1, \dots, \mathcal{D}\}$, is jointly strongly connected, and $\mathcal{G}(S_k) = \bigcup_{p=(k-1)B+1}^{kB} \mathcal{G}(\widetilde{\mathcal{A}}_p) = \mathcal{G}$, $\forall k \in \{1, \dots, \mathcal{D}\}$. Since $\mathcal{G}$ is strongly connected by hypothesis (Assumption II.2), it is possible to apply Theorem II.1. In particular, each of the $\mathcal{D}$ steps required to reach consensus is formed by B time instants (this property is necessary to ensure the condition of jointly strongly connectedness of each sequence S_k). Therefore, the entire process takes at most $B \cdot \mathcal{D}$ time steps, referred to the virtual time base. Given the equivalence stated in Lemma IV.1, the thesis is proved. □

B. Implementation Issues

Even though the convergence property of the asynchronous max-consensus protocol has been proved in the previous section, its detection relies on the knowledge of global quantities that are usually unknown to network nodes in applications (*e.g.,* the asynchronism measure B and the network diameter $\mathcal{D}$). In this section, we present an event-driven detection technique that is based only on the knowledge of local variables and guarantees the same upper bound on the convergence time of the previous time-driven approach. The proposed technique is inspired by [29] and [13], where it is used to solve a distributed connectivity control problem and a distributed target tracking application, respectively.

Each node possesses a binary vector of tokens $\mathbf{v}^{[i]} = [v_1^{[i]} \dots v_n^{[i]}]^T \in \{0,1\}^n$, $i \in \mathcal{I}$. This is a collection of n flags that are updated concurrently with the state of node i during the consensus process, on the basis of the token information received by its neighbors stored, similarly to the state variable, in a matrix of token vectors $\underline{\mathbf{V}}^{[i]} \in \{0,1\}^{n \times \mathcal{N}^{[i]}}$, whose columns are indicated as $\underline{\mathbf{v}}_j^{[i]} \in \{0,1\}^n$.

Before starting the consensus process, the token vector is set as

$$v_u^{[i]}(t_0^{[i]}) = \begin{cases} 1 & \text{if } u = i \\ 0 & \text{otherwise} \end{cases}, \forall i \in \mathcal{I}. \tag{14}$$

Notably, setting element $v_u^{[i]}$ to 1 indicates that node i has received the state value of node u. If condition $v_u^{[i]} = 1$ is verified for all $u \in \mathcal{I}$, then node i holds the maximum value. We observe that, for the implementation of this event-based convergence detection technique, the only global information needed by all nodes is the network size n. This information can be easily computed in a distributed way in order to keep the detection technique totally distributed [30].

Theorem IV.2. *If every node $i \in \mathcal{I}$ updates its token vector $\mathbf{v}^{[i]}(t_{k+1}^{[i]})$ at time $t_{k+1}^{[i]} \in \mathcal{T}^{[i]}$, according to the rule*

$$\mathbf{v}^{[i]}(t_{k+1}^{[i]}) = \bigvee_{j \in \mathcal{N}^{[i]}} \underline{\mathbf{v}}_j^{[i]}(t_k^{[i]}), \tag{15}$$

then condition $\bigwedge_{u \in \mathcal{I}} v_u^{[i]} = 1$, where operator $\bigwedge$ stands for logical and, *is verified after at most $B \cdot \mathcal{D}$ time units, expressed in the virtual time base (in (15), the* or *operator is intended element-wise, and for the token matrix update, the same rules in (4) can be applied). This condition ensures that the asynchronous consensus protocol correctly makes the system converge to the maximum value of the initial states $x^{[i]}(t_0^{[i]})$, $i \in \mathcal{I}$.*

Proof. The proof of convergence with the token-based mechanism follows the same line of arguments used in the proof of Theorem IV.1. Recalling Proposition IV.1, Eq. (14) can be rewritten as:

$$v_u^{[i]}(\tau_{0_i}) = \begin{cases} 1 & \text{if } u = i \\ 0 & \text{otherwise} \end{cases}. \tag{16}$$

Assuming partial asynchronism to hold, after at most B virtual time units node i has surely received the states of its neighbors $j \in \mathcal{N}^{[i]}$, and has updated its state. This implies that at virtual time τ_{0_i+B} the token vector is

$$v_u^{[i]}(\tau_{0_i+B}) = \begin{cases} 1 & \text{if } u \in \mathcal{N}^{[i]} \\ 0 & \text{otherwise} \end{cases}. \tag{17}$$

After further B virtual time units the token vector is

$$v_u^{[i]}(\tau_{0_i+2B}) = \begin{cases} 1 & \text{if } u \in \mathcal{N}^{[i]} \cup \bigcup_{p \in \mathcal{N}^{[i]}} \{\mathcal{N}^{[p]}\} \\ 0 & \text{otherwise} \end{cases}. \tag{18}$$

In other words, in $2B$ virtual time units, a path of length 2 has been established between node i and its 2-hop neighbors $w \in \bigcup_{p \in \mathcal{N}^{[i]}} \{\mathcal{N}^{[p]}\}$. Iteratively, after kB virtual time units, each node is able to establish a communication path with its k-hop neighbors, so that the relation

$$\mathbf{v}^{[i]}(\tau_{0_i+kB}) = \bigvee_{j \in \mathcal{N}^{[i]}} \mathbf{v}^{[j]}(\tau_{0_i+(k-1)B}) \tag{19}$$

holds. Thus, element u of the token vector at time kB can be inductively computed as

$$v_u^{[i]}(\tau_{0_i+kB}) = \begin{cases} 1 & \text{if } u \in \{\mathcal{N}^{[l]} | v_l^{[i]}(\tau_{0_i+(k-1)B}) = 1, l \in \mathcal{I}\}, \\ 0 & \text{otherwise.} \end{cases} \tag{20}$$

Since the network diameter $\mathcal{D}$ is the maximum length of the shortest path between any pair of nodes, at most $B \cdot \mathcal{D}$ virtual time units are required to obtain an all-one token vector $\mathbf{v}^{[i]} = [1 \ 1 \ \dots \ 1]^T$, $\forall i \in \mathcal{I}$. Therefore, the token mechanism ensures that every network node receives the state information from all the other nodes in the network in finite time. The convergence time of the procedure is bounded by the asynchronous max-consensus convergence result (Theorem IV.1), while the correctness of the convergence value is ensured by the associative property of the max-operator. $\square$

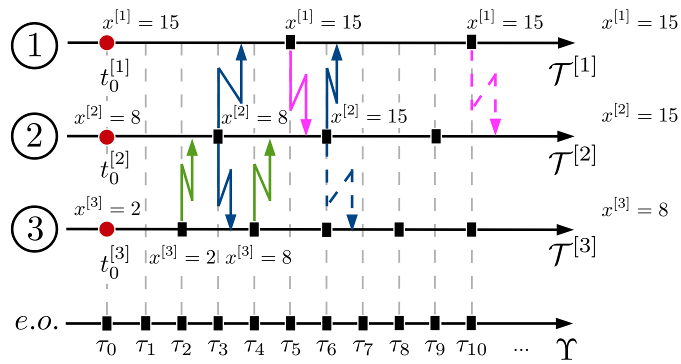


Fig. 2. Application of a synchronous max-consensus protocol in an asynchronous framework. Ticks indicate a max-consensus update for node i. Arrows indicate directed communication; dashed arrows indicate a communication that is not effective for the receiver node. Beside each update, the current state of the corresponding node is indicated. The *e.o.* timeline illustrates the point of view of an external observer.

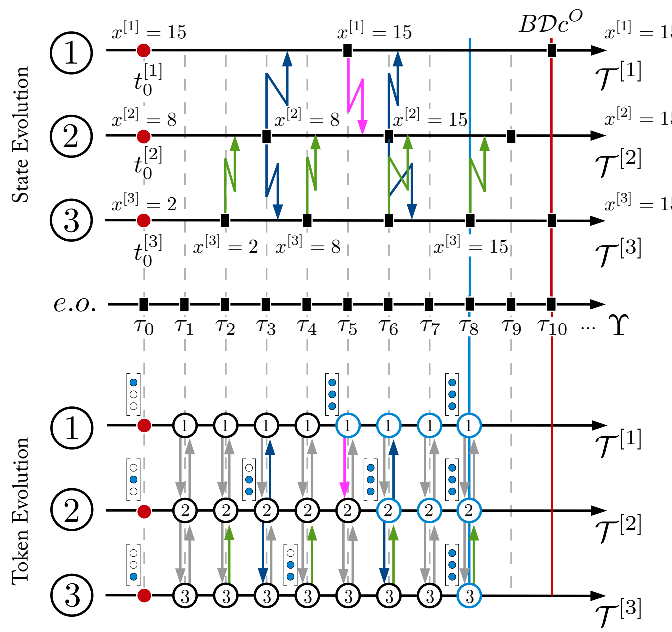


Fig. 3. Application of the asynchronous max-consensus protocol in an asynchronous framework. Evolution of the network states and of the token vectors for a network of 3 nodes (connected as in Fig. 1(a)) operating an asynchronous max-consensus with token-based convergence detection mechanism (the initial state values are: $x^{[1]}(t_0^{[1]}) = 15$, $x^{[2]}(t_0^{[2]}) = 8$, $x^{[3]}(t_0^{[3]}) = 2$). The vertical blue line illustrates the actual convergence time, while its upper bound is indicated by the red line.

V. Numerical Results

In this section, we provide an illustrative example that validates the theoretical results and the effectiveness of the proposed token-based convergence detection mechanism.

Consider a network of $n = 3$ nodes, connected in a line topology as the one previously reported in Fig. 1(a). At first, we assume that the network diameter, that is $\mathcal{D} = 2$ in this example, is known by all nodes. First, we highlight the issues connected to the erroneous application of the synchronous max-consensus protocol in an asynchronous framework. To this aim, we suppose that node $i \in \mathcal{I}$ runs the max-consensus protocol in Eq. (2) at each time instant $t_k^{[i]}$, $k \in \{1, 2\}$, of its

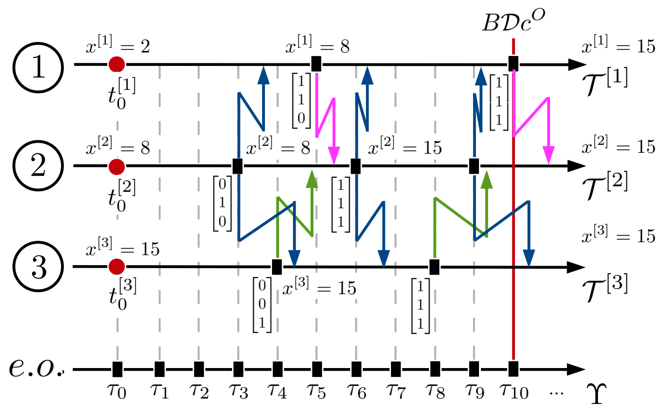


Fig. 4. Evolution of the state variables for a network of 3 nodes (connected as in Fig. 1(a)) operating an asynchronous max-consensus with token-based convergence detection mechanism, starting from the set of initial conditions $x^{[1]}(t_0^{[1]}) = 2$, $x^{[2]}(t_0^{[2]}) = 8$, and $x^{[3]}(t_0^{[3]}) = 15$. The red line indicates the upper bound on the convergence time. Here, convergence occurs exactly after $B \cdot \mathcal{D}$ time units.

local time base, assuming to work in a synchronous setting. In this framework, each node is aware that if the protocol is run for exactly $\mathcal{D}$ updates, then its state converge (see Theorem II.1) to the maximum value [10]. The initial state values of nodes are: $x^{[1]}(t_0^{[1]}) = 15$, $x^{[2]}(t_0^{[2]}) = 8$, $x^{[3]}(t_0^{[3]}) = 2$, respectively assumed at times $t_0^{[i]}$, $i = 1, 2, 3$ (see Fig. 2). Thus, the maximum value to which the network is supposed to converge is $x^{[1]} = 15$. We assume the following constant clock periods, $c^{[1]} = 0.5$ ms, $c^{[2]} = 0.3$ ms, and $c^{[3]} = 0.2$ ms, a maximum delay $\check{d} = 0.095$ ms, and all the starting times as synchronized. From the state evolution illustrated in Fig. 2, we observe that node 3 does not converge to the maximum value at the end of the execution of the max-consensus protocol. The state trajectories illustrated in Fig. 5(a) show that the presence of time misalignments and bounded delays do not yield global consensus when the synchronous protocol is used.

On the other hand, Fig. 3 illustrates the state evolution of the network nodes when the asynchronous protocol is correctly applied. We consider the same network with the same initial state values, clock periods, maximum delay, and synchronized starting times. It is straightforward to verify that the system is partially asynchronous with asynchronism measure $B = 5$ and $c^O = 0.1$ ms. In Fig. 3, the temporal sequence of sent/received messages, operated according to the asynchronous max-consensus protocol in Eq. (3) with token-based convergence detection in Eq. (15), is illustrated. The value of the information state is also illustrated in correspondence to each time instant in which it evolves to a new value (Fig. 5(b) illustrates the state trajectories of each node). We observe that in this case all nodes agree on the maximum value of the initial states (*i.e.*, consensus is reached) after 8 virtual time units (node 3 is the last one that updates its state value at $\tau_8 = 0.8$ ms). We remind that, in this case, the upper bound on the convergence time is $B \cdot \mathcal{D} \cdot c^O = 1$ ms. Thus, the token-based mechanism allows the nodes to detect the convergence condition faster than the time-driven solution given by Theorem IV.1.

Figure 4 illustrates a different scenario for the same network. This set-up differs from the previous one in the value of the clock period of node 3, $c^{[3]} = 0.4$ ms, and in the node's initial values, $x^{[1]}(t_0^{[1]}) = 2$, $x^{[2]}(t_0^{[2]}) = 8$, and $x^{[3]}(t_0^{[3]}) = 15$. Moreover, we emphasize the effects of time varying delays increasing the maximum delay to $\check{d} = 0.155$ ms. Also in this network, it results that $B = 5$ and $c^O = 0.1$ ms. In this case, the evolution of the token-based mechanism yields the network to convergence in exactly $B \cdot \mathcal{D} \cdot c^O = 1$ ms.

VI. Conclusions

In this paper, we have studied the convergence properties of the asynchronous discrete-time max-consensus protocol over static directed networks. The work is motivated by the need to implement this protocol in applications, where asynchronous updates and communication delays are usually present. The analysis is grounded on the equivalence between the asynchronous protocol running on a static network and a corresponding synchronous model running on a suitable switching topology. We have proved that, in presence of arbitrary asynchronous updates and bounded delays, the max-consensus protocol yields the convergence of a distributed system in finite time if the underlying communication network is strongly connected, under the mild hypothesis of partial asynchronism. Moreover, the convergence time is bounded by a quantity that depends on the network diameter and on the asynchronism measure. A fully distributed mechanism based on token vectors has then been implemented to detect convergence in real applications. It is proved that, in the worst case scenario, the token-based mechanism has the same upper bound on the convergence time as the one derived in the time-based framework. Finally, its effectiveness and efficiency is shown through a numerical example.

References

[1] Z. Li, Z. Duan, G. Chen, and L. Huang, "Consensus of multiagent systems and synchronization of complex networks: a unified viewpoint," *IEEE Transactions on Circuits and Systems I*, vol. 57, no. 1, pp. 213–224, January 2010.

[2] W. Ren, "Averaging algorithms and consensus," in *Encyclopedia of Systems and Control*, J. Baillieul and T. Samad, Eds. London: Springer, 2014, pp. 1–10.

[3] Z. Cheng, H.-T. Zhang, M.-C. Fan, and G. Chen, "Distributed consensus of multi-agent systems with input constraints: A model predictive control approach," *IEEE Transactions on Circuits and Systems I*, vol. 62, no. 3, pp. 825–834, March 2015.

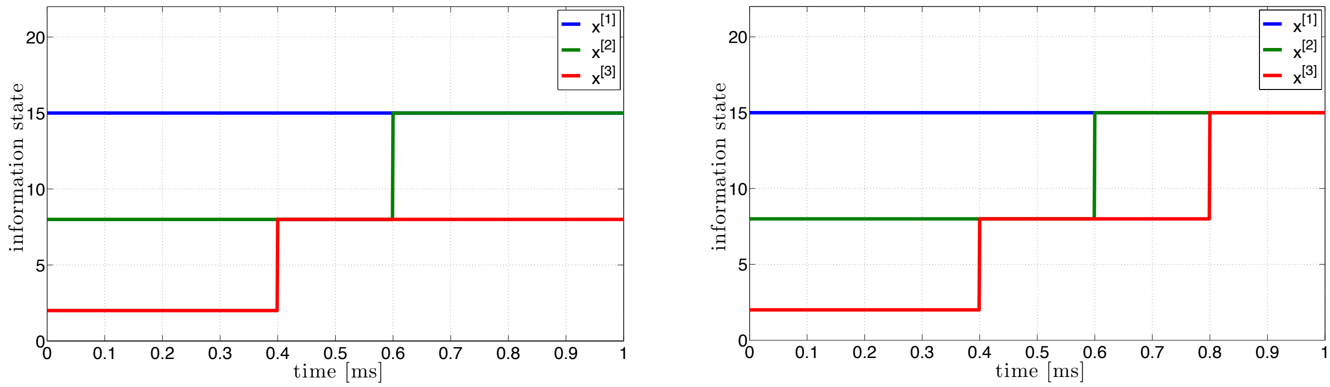
[4] J. Xi, Y. Yu, G. Liu, and Y. Zhong, "Guaranteed-cost consensus for singular multi-agent systems with switching topologies," *IEEE Transactions on Circuits and Systems I*, vol. 61, no. 5, pp. 1531–1542, May 2014.

[5] R. Olfati-Saber, E. Franco, E. Frazzoli, and J. Shamma, "Belief consensus and distributed hypothesis testing in sensor networks," in *Workshop on Networked Embedded Sensing and Control (NESC'05)*, Notre Dame, IN, USA, October 2005, pp. 169–182.

[6] Y. Yang, D. V. Dimarogonas, and X. Hu, "Opinion consensus of modified hegselmann-krause models," *Automatica*, vol. 50, no. 2, pp. 622–627, December 2014.

[7] D. Di Paola, A. Petitti, and A. Rizzo, "Distributed kalman filtering via node selection in heterogeneous sensor networks," *International Journal of Systems Science*, vol. 46, no. 14, pp. 2572–2583, January 2015. [Online]. Available: http://dx.doi.org/10.1080/00207721.2013.873836

[8] Z.-H. Guan, F.-L. Sun, Y.-W. Wang, and T. Li, "Finite-time consensus for leader-following second-order multi-agent networks," *IEEE Transactions on Circuits and Systems I*, vol. 59, no. 11, pp. 2646–2654, November 2012.



(a) (b)

Fig. 5. State evolution obtained through the application of (a) a synchronous max-consensus protocol (refer to Fig. 2) and (b) an asynchronous one (refer to Fig. 3) to the asynchronous network in Fig. 1(a) (node update times are illustrated, respectively, in Fig. 2 and 3).

[9] W. Ren, "Consensus strategies for cooperative control of vehicle formations," *IET Control Theory & Applications*, vol. 1, no. 2, pp. 505–512, March 2007.

[10] B. M. Nejad, S. A. Attia, and J. Raisch, "Max-consensus in a max-plus algebraic setting: The case of fixed communication topologies," in *International Symposium on Information, Communication and Automation Technologies (ICAT)*, Sarajevo, Bosnia and Herzegovina, October 2009, pp. 1–7.

[11] T. Borsche and S. A. Attia, "On leader election in multi-agent control systems," in *IEEE Chinese Control and Decision Conference (CCDC)*, Xuzhou, China, May 2010, pp. 102–107.

[12] T. Wongpiromsarn, K. You, and L. Xie, "A consensus approach to the assignment problem: Application to mobile sensor dispatch," in *IEEE International Conference on Control and Automation (ICCA)*, Xiamen, China, June 2010, pp. 2024–2029.

[13] S. Giannini, A. Petitti, D. Di Paola, and A. Rizzo, "Asynchronous consensus-based distributed target tracking," in *IEEE Conference on Decision and Control (CDC)*, Firenze, Italy, December 2013, pp. 2006–2011.

[14] S. Giannini, D. Di Paola, and A. Rizzo, "Coverage-aware distributed target tracking for mobile sensor networks," in *IEEE Conference on Decision and Control (CDC)*, Maui, HW, USA, December 2012, pp. 1386–1391.

[15] A. Petitti, D. Di Paola, A. Rizzo, and G. Cicirelli, "Consensus-based distributed estimation for target tracking in heterogeneous sensor networks," in *IEEE Conference on Decision and Control and European Control Conference (CDC-ECC)*, Orlando, FL, USA, December 2011, pp. 6648–6653.

[16] Z. Dengchang, A. Zhulin, and X. Yongjun, "Time synchronization in wireless sensor networks using max and average consensus protocol," *International Journal of Distributed Sensor Networks*, vol. 2013, pp. 1–10, 2013. [Online]. Available: http://dx.doi.org/10.1155/2013/192128

[17] L. Li, D. Ho, and J. Lu, "A unified approach to practical consensus with quantized data and time delay," *IEEE Transactions on Circuits and Systems I*, vol. 60, no. 10, pp. 2668–2678, October 2013.

[18] S. Giannini, D. Di Paola, A. Petitti, and A. Rizzo, "On the convergence of the max-consensus protocol with asynchronous updates," in *IEEE Conference on Decision and Control (CDC)*, Firenze, Italy, December 2013, pp. 2605–2610.

[19] Y. G. Sun, L. Wang, and G. Xie, "Average consensus in networks of dynamic agents with switching topologies and multiple time-varying delays," *Systems & Control Letters*, vol. 57, no. 2, pp. 175–183, February 2008.

[20] J. Zhan and X. Li, "Asynchronous consensus of multiple double-integrator agents with arbitrary sampling intervals and communication delays," *IEEE Transactions on Circuits and Systems I*, vol. 62, no. 9, pp. 2301–2311, Sept 2015.

[21] M. Cao, A. S. Morse, and B. D. Anderson, "Agreeing asynchronously," *IEEE Transactions on Automatic Control*, vol. 53, no. 8, pp. 1826–1838, September 2008.

[22] B. M. Nejad, S. A. Attia, and J. Raisch, "Max-consensus in a max-plus algebraic setting: The case of switching communication topologies," in *International Workshop on Discrete Event Systems - Part I*, Berlin, Germany, August 2010, pp. 173–180.

[23] F. Iutzeler, P. Ciblat, and J. Jakubowicz, "Analysis of max-consensus algorithms in wireless channels," *IEEE Transactions on Signal Processing*, vol. 60, no. 11, pp. 6103–6107, November 2012.

[24] D. Bertsekas and J. Tsitsiklis, *Parallel and Distributed Computation: Numerical Methods*. Belmont, MA, USA: Athena Scientific, 1997.

[25] J. Lin, A. Morse, and B. Anderson, "The multi-agent rendezvous problem. part 2: The asynchronous case," *SIAM Journal on Control and Optimization*, vol. 46, no. 6, pp. 2120–2147, November 2007.

[26] J. G. Manathara, A. Dukkipati, and D. Ghose, "Tropical algebraic approach to consensus over networks," *arXiv preprint arXiv:1109.0418*, pp. 1–11, September 2011.

[27] J. Bondy and U. Murty, *Graph Theory*, 3rd ed., ser. Graduate texts in mathematics. New York, NY, USA: Springer, 2008, vol. 244.

[28] F. Xiao and L. Wang, "Asynchronous consensus in continuous-time multi-agent systems with switching topology and time-varying delays," *IEEE Transactions on Automatic Control*, vol. 53, no. 8, pp. 1804–1816, September 2008.

[29] M. Zavlanos and G. Pappas, "Distributed connectivity control of mobile networks," *IEEE Transactions on Robotics*, vol. 24, no. 6, pp. 1416–1428, December 2008.

[30] G. Tel, *Introduction to Distributed Algorithms*, 2nd ed. New York, NY, USA: Cambridge University Press, 2000.

APPENDIX

Proof of Proposition IV.1. Consider node $i \in \mathcal{I}$ and its state variable $x^{[i]}(t_k^{[i]})$ at the update time $t_k^{[i]} \in \mathcal{T}^{[i]}$, $k \in \mathbb{N}$. Hence, assume:

$$\widetilde{x}^{[i]}(\tau_{k_i}) = x^{[i]}(t_k^{[i]}), \quad k \in \mathbb{N}, \tag{21}$$

where, from Eq. (7), $\tau_{k_i} = f^{[i]}(t_k^{[i]})$ (the equivalence $\widetilde{x}^{[i]}(\tau_{0_i}) = x^{[i]}(t_0^{[i]})$, $\forall i \in \mathcal{I}$, holds). We now prove that the application of a single step of the asynchronous max-consensus protocol $\mathcal{P}$, expressed by Eq. (3) and here rewritten as

$$x^{[i]}(t_{k+1}^{[i]}) = \max\{x^{[i]}(t_k^{[i]}), \max_{\substack{j \in \mathcal{N}^{[i]} \\ j \neq i}} \{\underline{x}_j^{[i]}(t_k^{[i]})\}\}, \tag{22}$$

takes the state variable $x^{[i]}$ to a value equal to the same value obtained for $\widetilde{x}^{[i]}$ through the application of the synchronous protocol (12) at every virtual update time occurring in the interval $[\tau_{k_i}, \tau_{(k+1)_i})$.

First, consider the case $\tau_{(k+1)_i} \neq \tau_{k_i} + c^O$, *i.e.*, $\tau_{k_i} + c^O$ does not refer to an update time for node i. It follows from Eq. (12-(ii)) that

$$\widetilde{x}^{[i]}(\tau_l) = \widetilde{x}^{[i]}(\tau_{k_i}), \quad \forall \tau_l \in \Upsilon \text{ s.t. } k_i \leq l < (k+1)_i. \tag{23}$$

Suppose now that, for some $j \in \mathcal{N}^{[i]}$, $j \neq i$, there exists $t_k^{ij} \in \mathbb{R}$, such that: (i) $\tau_{k_i} \leq t_k^{ij} < \tau_{(k+1)_i}$; (ii) $t_k^{ij} - d_k^{ij} = \tau_{q_j}$, where τ_{q_j} is the virtual time corresponding to the local time $t_q^{[j]} \in \mathcal{T}^{[j]}$, $q \in \mathbb{N}_0$, at which the most recent state value sent by node j has been received by node i. We indicate with $k^{ij} = \left\lceil \frac{t_k^{ij}}{c^O} \right\rceil$ the index of the virtual time base at which node i detects that the most-recent value has been received from j ($\lceil \cdot \rceil$ indicates the ceiling function). Therefore, for each $\tau_l \in \Upsilon$, $k^{ij} \leq l < (k+1)_i$, node i assumes that

$$\underline{\widetilde{x}}_j^{[i]}(\tau_l) = \widetilde{x}^{[j]}(\tau_{q_j}). \tag{24}$$

Assume now $\tau_h = \tau_{(k+1)_i} - c^O$, then in τ_{h+1}, applying Eqs. (10-(i)), (21), (23), the relation

$$\begin{aligned} \widetilde{x}^{[i]}(\tau_{h+1}) &= \max_{j \in \mathcal{N}^{[i]}} \{\underline{\widetilde{x}}_j^{[i]}(\tau_h)\} = \\ &= \max\{\widetilde{x}^{[i]}(\tau_{k_i}), \max_{\substack{j \in \mathcal{N}^{[i]} \\ j \neq i}} \{\underline{\widetilde{x}}_j^{[i]}(\tau_{k^{ij}})\} \end{aligned} \tag{25}$$

holds. By comparison of Eqs. (22) and (25), it is straightforward to verify that $x^{[i]}(t_{k+1}^{[i]}) = \widetilde{x}^{[i]}(\tau_{(k+1)_i})$. $\square$